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Course - DBMS

Assignment -1.

Section- A:-

Ques1. what is DBMS? why it is required?
Ans A DBMS(Database Management system) is a software that store, manage and retrieve data in an organized way it is required to use data redundancy maintain consistency. Provide security and allow easy data access.

Ques2. Why is DBMS better than traditional file system?
Ans DBMS reduces duplication of data, avoids inconsistency provides security, support multiple users at the same time and offers backup and recovery facilities.

Ques3. What is Data Abstraction? Explain its levels?
Ans Data Abstraction means hiding internal details and showing only necessary information.
There levels are:
-> Physical level - How data is stored
-> Logical level - What data is stored and relationships
-> view level -> How use see the data

Ques4. What is 3 schema Architecture?

Ans Three schema architecture includes internal, conceptual and external level it is important because it provides data independence and seperates views from physical storage.

Ques5. What is data independence? Why is it needed?

Ans Data independence means changes at one level of data base do not affect other levels. It is needed for flexibility and easy modification of the system.

Ques6. What is an ER model?

Ans ER model is conceptual diagram that represents entities and consistency of data examples are primary key, foreign key, unique and NOT NULL constraints.

~~Section B: Design~~

Ques7. What are Integrity constraints?

Ans Integrity contraints are rules that maintain accuracy and consistency of data Examples are primary key, foreign key, unique and NOT NULL constraints.

Section B Design, Draw & Defend

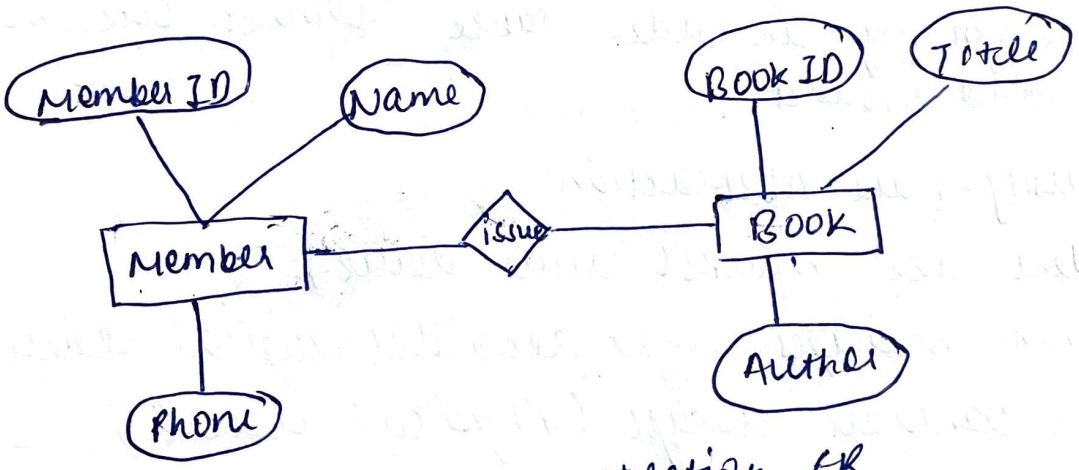
Three - Schema

External level → user view
↓
Conceptual level → overall structure
↓
Internal level → physical storage.

ii) Library ER Diagram

- BOOK (BOOK ID, Title, Author)
- Member (Member ID, Name, Phone)

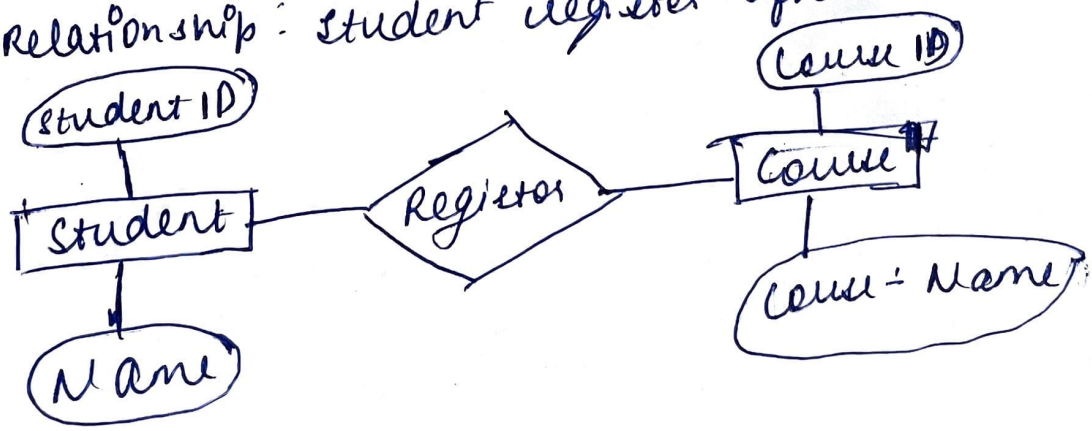
Relationship : Member issue Book



iii) Student - course Registration ER

- Student (Student ID, name)
- Course (Course ID, Course Name)

Relationship : Student register after course



section-C Case Base Analytical question

i) ER-diagram Hospital system

Entities : Patients, Doctors, Appointment

Patients book Appointment.

Doctor attends Appointment.

Primary key : Patient ID, Doctor ID, Appointment ID

ii) online shopping

Customer (Customer ID, - PK)

Product (Product-ID - PK)

Order (order -ID - PK)

Foreign key in order Table connect customer and Product

iii) University - Data Abstraction

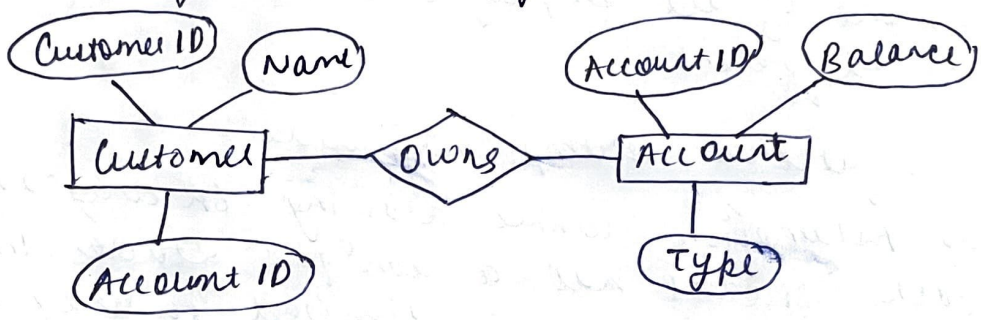
Student see marks (view level)

Admin manages ~~age~~ records (logical level)

DBA controls storage (Physical level) -

section D : One Diagram + One Insight

i) ER diagram – ATM System



Acct_NO is Primary key
Relation : Customer Owns Account.

Account No it take uniquely intentifier Account

ii) Three schema Architecture + one Benefit

Benefit: Provides Data Independence –
Changes in storage do not affect user.

Diagram: External level (user view)
↓
conceptual level (logical structure of Database)
↓
Internal level (Physical storage level)

iii) Draw a entity with Attributes

Entity : Employee
Employee_ID (PK)
Name
Salary
Department.

Primary key : employee_ID
because of all employee their ID must be different.

Section E : Reflection Assumption & Wha

i) Two Assumption while drawing ER diagram

Ans
- Each student has a unique student ID
- One Appointment is limited to only on doctor

ii) one Limit of ER modeling
- Cannot represent complex constraints easily
- Does not show procedural aspects
- Not suitable for ~~p~~ large dynamic system directly.

iii) Most Difficult Part in Unit 1

The most difficult part was understanding, data independence, because it requires clear understanding of 3 schema architecture and how level interact